Marawan Eldeib

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Summary

Software engineer and M.Sc. Information Technology student working on secure system design and LLM-based AI systems. Optimised NASA/DLR research software at the Institute of Space Systems, measured how LLM-integrated tools leak confidential data, and built and hardened a production full-stack platform.

Education

M.Sc. Information Technology (Specialisation: Software Engineering) **Oct 2024 – Mar 2027 (expected)**

University of Stuttgart Stuttgart, Germany

Relevant coursework: Deep Learning Lab, Distributed Systems, Service Management and Cloud Computing, Research Project (LLM data exposure).

B.Eng. (Hons.) Electronics Majoring in Computer **Jul 2020 – Oct 2024**

Multimedia University Cyberjaya, Malaysia

CGPA: 3.7/4.0 — First Class Honours (German equivalent ≈ 1.4).

Work Experience

Student Research Assistant (HiWi) **Oct 2026 – Present**

University of Stuttgart, Institute for Control Engineering of Machine Tools (ISW) Stuttgart, Germany

- **Research software:** supporting the INTERFACE4CPPS research project (ISW with KASTEL at KIT), which develops a software prototype that integrates engineering artifacts and data exchange formats for cyber-physical production systems.

Working Student – Software Engineer **Apr 2026 – Aug 2026**

University of Stuttgart, Institute of Space Systems (IRS), SOFIA Data Center Stuttgart, Germany

- **Performance optimisation:** took a NASA/DLR data-reduction job that could not complete within available memory down to a 14.4 GB peak, without changing the output.
- **Root-cause analysis:** traced the peak to redundant concurrent calculations; a two-line config change gave 39%, and memray-guided dtype fixes a further 32%.

Working Student – Software Engineer **Apr 2025 – Dec 2025**

Fraunhofer IOSB Karlsruhe, Germany

- **LLM evaluation:** tested and evaluated Java and Spring Boot backend services for LLM-based data extraction across Asset Administration Shell (AAS) digital-twin applications, benchmarking six large language models.
- **Test automation:** developed automated end-to-end test suites and curated 30+ evaluation datasets to validate model performance.

Intern – Software Engineer **Jul 2023 – Oct 2023**

Ikhlas Com Sdn Bhd (AirAsia Group) Kuala Lumpur, Malaysia

- **Feature delivery:** resolved 14 defects and shipped 5 checkout-calculation features, improving application stability for 10,000+ users.
- **Refactoring & APIs:** refactored 2 application modules and integrated GraphQL APIs for bilingual content management within Agile sprints.

Projects

Master's Thesis — A GitOps-Constrained Agentic Workflow for System Manipulation **Oct 2026 – Mar 2027**

LLM Agents | GitOps | DevSecOps | HPC | Kubernetes

Supervised by Prof. Dr. Marco Aiello (IAAS), in collaboration with the High Performance Computing Center Stuttgart (HLRS).

Secure GitOps workflow that limits LLM agents to reviewed pull requests on a shared HPC cluster, with code changes and job submission controlled through MCP gateways, sandboxed execution, audit trail and rollback; includes threat modelling and a security evaluation.

Measuring Data Exposure in LLM-Integrated Productivity Tools — Apr 2026 – Sep 2026

Python | mitmproxy

- **Network traffic analysis:** intercepted outbound HTTPS and WebSocket traffic with Python and mitmproxy, using planted canaries to trace every leaked fragment.
- **Findings:** writing-assistant extensions, Microsoft Word, Edge Copilot and desktop assistants each sent all 12 planted secrets to their servers without any sharing step (no-extension baseline: 0 of 12).

Saha — Secure Full-Stack Platform — Mar 2026 – Present

Next.js | PostgreSQL | TypeScript

- **Software & security engineering:** designed, built, and deployed a production full-stack booking platform secured with Postgres row-level security, a strict Content Security Policy, MFA-gated admin and audit logging, and conducted a self-audit that uncovered and remediated a privilege-escalation vulnerability.

KGFM — Foundation Models for Relational Databases — Nov 2025 – Feb 2026

PyTorch | PyTorch Geometric

- **Graph ML research:** conducted the first systematic comparison of random-walk (Flock) and message-passing (ULTRA/POSTRA) knowledge-graph foundation models for zero-shot transfer to relational-database tasks (RelBench F1); Flock reached 0.894 MRR on link prediction, a 17.6% zero-shot gain over ULTRA, with its zero-shot result exceeding ULTRA's fine-tuned performance.

MangoVision — Aerial Object Detection — Nov 2023 – Jul 2024

PyTorch | YOLOv8 | Faster R-CNN | Detectron2

- **Deep learning:** developed and benchmarked YOLOv8 and Faster R-CNN detectors across a public benchmark and a custom drone dataset, reaching mAP@0.5 of 0.959 and 0.756. *Best Research Project Award, Gold Medal.*

Publications

M. H. Lye and **M. A. Eldeib**, "Performance Analysis of Faster R-CNN and YOLOv8 Model for Mango Fruits Detection," *Journal of Informatics and Web Engineering*, vol. 5, no. 2, pp. 280–289, 2026. doi: doi.org/10.33093/jiwe.2026.5.2.17

Technical Skills

- **Security:** Web Application Penetration Testing, Burp Suite, OWASP ZAP, OWASP Top 10 (XSS, SQL injection, CSRF, CORS, OAuth 2.0), binary exploitation and exploit mitigation (self-study), SIEM and log analysis (Splunk, IBM QRadar).
- **Networks:** HTTP/HTTPS, TLS interception, network traffic analysis (mitmproxy, Wireshark), Nmap.
- **Cloud, DevOps & Tooling:** Docker, Git, GitHub Actions, Playwright, AWS (Lambda, API Gateway, DynamoDB, ELB, Auto Scaling), Kubernetes, Terraform, Ansible, ArgoCD, Jenkins.
- **Machine Learning & AI:** PyTorch, PyTorch Geometric, scikit-learn, Detectron2, pandas, NumPy, LLMs, Computer Vision, MCP, Claude Code, Model Evaluation & Benchmarking.
- **Programming:** Python, C/C++, Java, SQL, JavaScript, TypeScript, Flutter; profiling with memray.
- **Software Engineering:** React, Next.js, Node.js, Spring Boot, REST APIs, GraphQL, JSON, PostgreSQL, Redis, Tailwind CSS.
- **Linux & HPC:** Ubuntu, Kali, Bash, YAML, SSH, Slurm (GPU job submission).
- **Project Management:** Agile/Scrum, Jira, Confluence.

Languages: Arabic (Native) | English (C1) | German (B1).